Indicator	Unit	Baseline	Optimized	Change
GLOBAL WARMING (GWP)	kg CO_2 eq	3.1613 ± 0.2162	1.6965 ± 0.1543	-46.34%
ACIDIFICATION	kg SO_2 eq	0.0141 ± 0.0010	0.0067 ± 0.0006	-52.58%
WATER USAGE	m^3	0.0304 ± 0.0023	0.0210 ± 0.0021	-31.09%
FEEDSTOCK COST	USD	26.1375 ± 2.1721	6.7625 ± 0.1991	-74.13%

ANALYSIS & CHARTS

Uncertainty Propagation: Global Warming (Monte Carlo Normal Distribution PDF)

Optimization tradeoffs bar chart

Uncertainty distribution histogram

PROCESS SCALING

PROCESS EFFICIENCY 1.00

RECYCLE SUBSTITUTE RATE 0.00

LOSS / SCRAP FACTOR 0.00

DATABASE PARAMETER TUNING

-- Tune background feedstock --

TOPSIS MCDA WEIGHTS

CARBON (GWP) WEIGHT 40%

ACIDIFICATION WEIGHT 15%

WATER WEIGHT 15%

MATERIAL COST WEIGHT 30%

No database parameters loaded.

TVL MASS BALANCE

VERIFICATION STATUS PASSED

INPUTS	OUTPUTS
21.650 kg	21.650 kg

STOICHIOMETRIC BALANCE ERROR
0.0000%

ELEMENTAL BALANCE PASSED

✓ Elemental stoichiometry balanced

DATABASE DIAGNOSTICS

DATABASE HEALTH UNCHECKED

Scan Context

ENGINEERING JUSTIFICATION REPORT

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> *The substitution of virgin glass fibers with sorted glass cullet in the web-synthesized product manufacturing process for a Next-Gen Silicon Solar Cell Module represents a significant environmental and economic improvement aligned with the specified decision weights. The reduction in carbon footprint by 46.34% (Global Warming Potential: -46.34%, from 3.1613 to 1.6965 kg CO2 eq) is a critical outcome, particularly given that it carries a high priority weight of 40%. This decrease in greenhouse gas emissions is primarily due to the lower embodied carbon associated with glass cullet compared to virgin glass, as cullet contains recycled materials and thus has a significantly reduced production energy footprint. Additionally, the substantial reduction in terrestrial acidification potential by 52.58% (from 0.0141 to 0.0067 kg SO2 eq) with an associated weight of 15% further solidifies this substitution's alignment with user priorities. The chemical composition and processing requirements of cullet result in a lower generation of sulfur dioxide, a key component in acidification, compared to virgin glass. The water consumption footprint reduction of -31.09% (from 0.0304 to 0.0210 m^3) with a weight of 15% demonstrates an additional environmental benefit through the reduced strain on freshwater resources associated with cullet processing. Lastly, the substantial cost reduction of -74.13% (from 26.1375 to 6.7625 USD) with a priority weight of 30% underscores the economic viability and sustainability of this feedstock substitution. Utilizing cullet as a raw material not only decreases operational costs but also fosters a more circular economy by repurposing waste materials. Overall, this swap is Pareto-improving, decoupling environmental footprints (carbon and acidification) without increasing operational costs, thus meeting the user's prioritization criteria effectively. The underlying physical and chemical reasons for these benefits are rooted in the lower energy requirements, reduced emissions, and economic incentives associated with using recycled materials over virgin ones.*